7.3.2 Volume of a hollow cylinder (volume of the material used)

Let the internal and external radii of a hollow cylinder be r and R units respectively. If the height of the cylinder is h units then

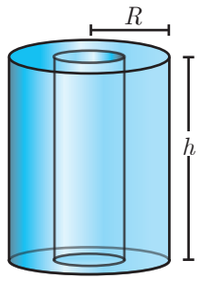


Fig. 7.26

The volume $V = \begin{Bmatrix}\text{volume of the}\\ \text{outer cylinder}\end{Bmatrix} - \begin{Bmatrix}\text{volume of the}\\ \text{inner cylinder}\end{Bmatrix}$

$$V = \pi R^2 h - \pi r^2 h = \pi(R^2 - r^2)h$$

Volume of a hollow cylinder $= \pi(R^2 - r^2)h$ cu. units.

Example 7.15 Find the volume of a cylinder whose height is 2 m and whose base area is 250 m^2.

Solution Let r and h be the radius and height of the cylinder respectively.

Given that, height $h = 2$ m, base area $= 250\ m^2$

Now, volume of a cylinder $= \pi r^2 h$ cu. units

$= base\ area \times h$

$= 250 \times 2 = 500\ m^3$

Therefore, volume of the cylinder $= 500\ m^3$

Thinking Corner

1. If the height is inversely proportional to the square of its radius, the volume of the cylinder is _____.
2. What happens to the volume of the cylinder with radius r and height h, when its (a) radius is halved (b) height is halved.

Example 7.16 The volume of a cylindrical water tank is 1.078×10^6 litres. If the diameter of the tank is 7 m, find its height.

Solution Let r and h be the radius and height of the cylinder respectively.

Given that, volume of the tank $= 1.078 \times 10^6 = 1078000$ litre

$= 1078\ m^3 \qquad (\because 1l = \frac{1}{1000}\ m^3)$

diameter $= 7m \Rightarrow$ radius $= \frac{7}{2}$ m

volume of the tank $= \pi r^2 h$ cu. units

$$1078 = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times h$$

Therefore, height of the tank is 28 m

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